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Adjective Ordering Across Languages

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Abstract

Adjective ordering preferences stand as perhaps one of the best candidates for a true linguistic universal: When multiple adjectives are strung together in service of modifying some noun, speakers of different languages—from English to Mandarin to Hebrew—exhibit robust and reliable preferences concerning the relative order of those adjectives. More importantly, despite the diversity of the languages investigated, the very same preferences surface over and over again. This tantalizing regularity has led to decades of research pursuing the source of these preferences. This article offers an overview of the findings and proposals that have resulted.

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1. INTRODUCTION

Confronted with a box that is brown and small, a speaker of English will describe the object as a *small brown box*; reversing the order of adjectives results in, at best, a cumbersome phrase: *brown small box*. But speakers of English are not the only ones who speak of small brown boxes; the very same preferences arise in Greek, Mandarin, and Tagalog, to name only very few of the languages claimed to prefer adjectives ordered in this way. These languages all feature prenominal adjectives that appear before the modified noun. In postnominal languages where adjectives follow the nouns they modify, we find the same English-like preferences, just in the reverse order. So, postnominal languages speak of a *box brown small*, like in Arabic, Hebrew, or Spanish. At issue appears to be the relative distance of adjectives to the nouns they modify, and the glaring question is why we should continue to find the same preferences in language after language. As with most discussions of linguistic universals, the tendency is to assume that a phenomenon so stable cross-linguistically must evidence some deep principle of the cognitive architecture shaping language. How else could we explain the emergence of similar preferences in unrelated languages?

To understand adjective ordering preferences, the work lies in first documenting the preferences to be explained, then in precisely specifying the relevant knowledge, and ultimately in identifying and understanding the pressures that shape it. As in most topics in linguistics, English has received much of the limelight, both empirically and theoretically, especially with early work or larger-scale quantitative investigations. Still, the main point of adjective ordering preferences is their cross-linguistic regularity, which means that observations beyond English make up an essential part of the literature on the topic. More importantly, a comprehensive theory of adjective ordering must account for the full range of languages in which these preferences arise.

This article reviews significant developments in the study of adjective ordering, from nineteenth-century grammatical descriptions (Sweet 1898) to latent-variable models of the twenty-first century (Leung et al. 2020). Limitations of space necessitate a brisk pace through this survey of 100+ years of research; readers interested in earlier reviews should consult Posner (1986) or Wulff (2003). We begin with a discussion of the various predictors that have been proposed to account for adjective ordering preferences. We then turn our sights to empirical investigations, seeing which predictors hold up to behavioral and corpus data. Having surveyed the predictors and evaluated their performance, we consider the various explanations offered for why a given predictor should play the role it does in adjective ordering. We close with a look ahead at future directions in investigations of adjective ordering.

2. ENUMERATING PREDICTORS

2.1. Laying the Groundwork

Much of the discussion of factors relevant to adjective ordering is previewed already in three short paragraphs from Sweet (1898). While some of Sweet's prescience might owe to the room for interpretation left by his pithy style, we shall see that the ideas continue to arise in the decades (and century) that followed.[1] He identifies three pressures that influence the relative order of multiple modifiers in English (he includes more than just attributive adjectives in his discussion). First, Sweet looks at adjective-noun pairs, concluding that properties of the meaning that results from those pairings influence the relative order of adjectives: "When a noun has more than one

[1] For those intrigued by strong personalities in the history of linguistics, the cantankerous Henry Sweet stands as one of the more entertaining. George Bernard Shaw names Sweet as his primary inspiration for the character of Professor Henry Higgins.

modifier, the general principle is that the one most closely connected with it in meaning comes next to it, as in *the three wise men*, where *wise men* is equivalent to the single word *sages*" (Sweet 1898, p. 8; emphasis added).

While closeness in meaning might not immediately suggest a concrete operationalization, the example Sweet provides (*wise men* ≈ *sages*) offers a useful clue. At issue appears to be whether the adjective-noun pairing forms a concept in its own right. This notion of concept formation/natural kinds/compounding/idiomaticity and its role in adjective ordering appears repeatedly in the literature: First you create the concept (e.g., *wise men*), then you modify that concept with further adjectives (Ziff 1960, Hetzron 1978, Byrne 1979, McNally & Boleda 2004, Bouchard 2005, Svenonius 2008).

Next, Sweet claims that properties of individual adjectives determine their placement, such that adjectives with a more specialized meaning appear closer to the noun: "Qualifiers come before such groups [of adjective-noun clusters closely connected in meaning], the one that is the most special in meaning (*three*) coming next to it. Hence there is a gradation of increasing specialization from the beginning to the end of such a group (*the*, *three*, *wise*)" (Sweet 1898, p. 8; emphasis added). Here again the reader must do some digging to uncover the intended operationalization. Unfortunately, the example given contains only one adjective (i.e., *wise*), and that adjective was meant to be handled by the principle of closeness in meaning above. Still, interpreting meaning specialization literally, it is the case—at least impressionistically—that *wise* is the most restrictive in terms of the nouns with which it may occur, while *the* is the least restrictive. Assuming restrictiveness to be the operative notion, we encounter this idea again from Ziff (1960), who claims that adjectives with a greater "privilege of occurrence" (essentially, how many different nouns an adjective may appear with) appear farther from the noun (see also Seiler 1978).

For his third principle, Sweet observes that both of the above considerations—closeness in meaning and meaning specialization—may be overridden in cases in which adjective-noun pairs form frequent collocations: "Very frequent collocations such as *old man*, *young man* (=*youth*) have become so fixed that no other adjective, even if more special in meaning, is allowed to come between the two words" (Sweet 1898, pp. 8–9; emphasis added).

Although Sweet presents collocation frequency as an independent factor, there are interesting parallels with closeness in meaning and concept formation: Adjective-noun pairings that name a concept or kind should appear together as a collocation more often than we would expect based on the frequencies of the individual words. Other authors have seized on similar ideas. (Martin 1969b, p. 473) talks of "association strength" between adjectives and nouns, as measured by how frequently a given adjective has modified a given noun; Ney (1983, p. 103) offers the intuitive label of "habit strength"; and Posner (1986, p. 330) describes this notion as "noun-specific frequency."[2]

After enumerating these principles, Sweet (1898, p. 9) closes by remarking that "when the modifiers are about equally balanced, the order may vary." While this may seem a small point, there is an important takeaway: Sweet leaves open the possibility that certain strings lack stable ordering regularities, thereby rejecting the notion that there exists a total order on adjective pairings.

2.2. Twentieth-Century Ideas

Fast-forward a few decades and the topic of adjective ordering gets taken up by Whorf (1945) (of Sapir–Whorf fame). Departing from Sweet, Whorf identifies two classes of English adjectives:

[2]Some authors have gone so far as to exclude specific adjectives—particularly *young*, *little*, and *old*—from analysis because of the adjectives' tendency to form frequent collocations (see Wulff 2003, p. 277, for discussion).

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those that refer to "inherent" qualities (e.g., color, material, provenance) and the "residuum" of adjectives referring to qualities not covered by the first class (e.g., size, shape, evaluation). According to Whorf, adjectives in the first class—the ones naming inherent qualities—appear nearer to the noun. Whorf's notion of inherentness evokes Behaghel's (1932, p. 4) first law: "Things belonging closely together in mind are also put closely together in communication." The notion also overlaps with the concept of intrinsicalness from Danks & Glucksberg (1971), and there are clear parallels with the individual- versus stage-level distinction or the temporariness of the properties involved (Carlson 1977, Larson 1998, Cinque 2010). Seiler (1978) links inherentness with the notion of meaning specificity: Less broadly applicable adjectives are more likely to name properties inherent to the modified noun.

Martin (1969a) advances the novel idea that "definiteness of denotation" determines ordering preferences in English. Although Martin names Sweet (1898) and Ziff (1960) as inspirations for this idea, it is not obvious that the concept maps neatly onto meaning specialization or privilege of occurrence. For Martin, definiteness of denotation operationalizes as the extent to which an adjective's meaning remains invariant to the specific noun it modifies. In other words, how much is an adjective's meaning independent from the context in which it is used? According to Martin, adjectives whose meaning varies wildly depending on the noun (compare *big mouse* with *big elephant*) are preferred farther from the noun than adjectives with a more definite meaning (compare *cardboard mouse* with *cardboard elephant*). While there is likely to be a relationship between definiteness of denotation and meaning specificity—adjectives occurring with a more definite meaning may be restricted to a smaller set of modifiable nouns—the more natural connection is to absolute versus relative properties (Sproat & Shih 1991) or to the semantic distinction between intersective and subsective modification (Truswell 2009). Indeed, Martin offers absoluteness as an alternative operationalization of his predictor, that is, which adjectives require fewer comparisons among candidate objects to determine whether the property holds.[3]

Quirk et al. (1972) advance the idea that adjectives naming more objective, visually observable properties get placed nearer to the noun, while "subjectively assessible" content appears farther away (see also Quirk et al. 1985). Hetzron (1978, p. 178) offers a streamlined version of the proposal along the dimension of subjectivity: "Place the more objective and undisputable qualifications closer to the noun, and the more subjective, opinionlike ones farther away." Four decades later, Scontras et al. (2017) return to the notion of subjectivity, arguing that it subsumes notions like meaning specificity, inherentness, or definiteness of denotation; the authors operationalize adjective subjectivity as the potential for faultless disagreement between a speaker and a listener about whether the relevant property holds of some object (for more on faultless disagreement, see Kölbel 2004, MacFarlane 2014).

So far, we have focused on predictors of adjective ordering that attend to aspects of an adjective's meaning. However, not all authors have looked at meaning as the source of ordering preferences; some have looked instead at syntactic or distributional properties. For example, Vendler (1963, 1968) and Annear (1964) propose that adjectives divide into classes on the basis of how they get introduced syntactically, with some adjectives requiring more elaborate syntactic structure than others; a series of transformations then yield what we see as multi-adjective strings. While there may be semantic correlates of the structural choices, ordering preferences are taken to arise on the basis of derivational syntactic proximity between nouns and adjectives in the pretransformed predicative structures.

[3] Sproat & Shih (1991, p. 593) explicitly reject Martin's (1969a) operationalization of absoluteness.

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As part of a more general theory of the role of perception in language processing, Bever (1970, p. 325) proposes that "in a series of prenominal adjectives, the more nounlike adjectives are ordered to be closer to the head noun they modify." Thus, colors (*red*) and materials (*plastic*), which readily serve as both adjectives and nouns, are placed nearer to the head noun than sizes (*large*), which rarely find uses as nouns. Indeed, this notion of nounlikeness and its role in adjective ordering is present already in Vendler (1963, p. 456), although for Vendler the operative factor is an adjective's syntactic connection with the modified noun, and nounlikeness merely correlates with that property. Biber et al. (1999, p. 599) offer a similar observation, namely that the "most nounlike" modifiers tend to occur closest to the head noun. However, for these authors, nounlikeness is a byproduct of adjective meaning.

In addition to meaning-based and distributional factors, two lower-level properties of adjectives have been argued to influence their position in multi-adjective strings. Martin (1969a) shows adjective frequency to play a role in ordering preferences, such that more frequent adjectives occur farther from the noun (i.e., first in the linear sequence in English; see also Wulff 2003). Hetzron (1978, p. 176), citing Behaghel's (1932) law of increasing terms, argues that adjective length also plays a role in ordering preferences, such that languages exhibit a "euphonic tendency" to place short adjectives before longer ones.

2.3. Class-Based Descriptions

In addition to psychological factors meant to predict preferences, the literature on adjective ordering is rife with descriptive accounts that map, as it were, the empirically observed preferences. Common to these accounts are a reliance on lexical-semantic classes of adjectives and an ordering thereof.

Dixon (1977) pioneers the approach in his systematic accounting of the "semantic types" lexicalized as adjectives in English (see also Brown 1965, p. 282). Guided by the hypothesis that such types are universal cross-linguistically, Dixon sets out to document the variation in syntactic properties associated with each type among 17 languages. Dixon identifies seven semantic types, or lexical-semantic classes of adjectives.[4] In the process, he provides a thorough description of the English system, including, crucially, a description of ordering preferences for unmarked contexts. According to Dixon, the adjective classes are ordered as follows, with higher-ranked classes appearing farther from the noun:

value > dimension > physical property > speed > human propensity > age > color.[5]

While Dixon's ordering claims are focused on English, he reports on descriptions of Hungarian and Telugu consistent with the English preferences (Dixon 1977, pp. 75–76). In an updated version of the paper, Dixon (1982, p. 26) additionally mentions Selepet, a language with postnominal adjectives in which the English order is mostly preserved in the mirror image.

In an attempt to codify such descriptive class-based orderings in the syntax, Cinque (1994) argues for an articulated nominal structure in which different classes of adjectives inhabit distinct

[4]Dixon (1977, p. 74) acknowledges that position adjectives (e.g., *high*, *left*, *near*) are omitted from the analysis as well as "a few other adjectives which pose particular difficulties." Already we see a rather serious problem with class-based accounts: Not all adjectives readily submit to classification.

[5]Dixon (1977, p. 37) also mentions two classes of "postadjectival modifiers," those describing origin/composition (*oatmeal* in *oatmeal dog food*) and those describing purpose/beneficiary (*dog* in *oatmeal dog food*); these classes occur nearest to the noun.

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syntactic functional projections. In this structure, a given adjective will appear as a specifier to the functional head handling its class, so that, for example, a size adjective like *big* inhabits the specifier position of XP (i.e., the phrase headed by the abstract functional element X), while a color adjective like *brown* serves as the specifier of YP; crucially, the relevant functional projections are ordered hierarchically (i.e., XP > YP), thereby delivering a fixed linear order. Cinque offers several arguments in favor of this architecture, all of them relying on the overarching motivation to grammatically encode ordering preferences as ordering restrictions, such that the grammar of a language can only generate the "unmarked serializations" (Cinque 1994, p. 95). With a rigid functional structure, only orderings that conform with that structure are possible.

Given that Cinque focuses on the architecture with which to encode ordering regularities within DP, he does not linger on the specific lexical classes or the adjectives that populate them. Still, drawing on data from Italian, he does provide some suggestions. First, he treats event nominals (e.g., *reazioni* 'reactions') and object-denoting nominals (e.g., *quadri* 'paintings') separately. For event nominals, he proposes the following ordering:

speaker-oriented > subject-oriented > manner > thematic.

For object-denoting nominals, Cinque follows the description by Sproat & Shih (1991), which shares many similarities with Dixon's (1977) hierarchy above:

quality > size > shape > color > nationality.

Building on Cinque's architectural assumptions, Scott (2002) makes explicit many of the claims that are implicit for Cinque, particularly the idea that cross-linguistic ordering regularities arise as a result of their being "part of Universal Grammar" (Scott 2002, p. 91).[6] In other words, according to Scott, we find the same orderings across languages because the functional structure delivering those orderings comes encoded in our genetic endowment.[7] Scott also spends more time fleshing out the details of that functional structure for "simple and result nominals" (e.g., *dog*, *exam*; Scott 2002, p. 112), arriving at the following proposed hierarchy:

subjective comment > ?evidential > size > length > height > speed > ?depth > width > weight > temperature > ?wetness > age > shape > color > nationality/origin > material > compound element.

To enforce a rigid ordering among classes, the full functional structure is always projected, regardless of which adjectives are present. In other words, every nominal always contains its full retinue of functional projections.

One puzzle that immediately presents itself is how to deal with adjective ordering in languages with postnominal adjectives. For Cinque (1994), this puzzle serves as part of the motivation for his architecture, at least when viewed through the lens of ANA languages, which have both pre- and postnominal adjectives. According to Cinque, ANA orders may be derived via leftward N movement in an underlying AAN structure; with an abundance of functional structure between N and D, there are plenty of landing sites for the movement to target. Moreover, according to

[6] To be fair, Cinque himself makes many of these assumptions explicit in his follow-up work on adverb ordering (Cinque 1999), from which Scott (2002) also draws inspiration.

[7] Scott's strong claims regarding the contents of Universal Grammar are considerably weakened when he allows for the possibility that the ordering hierarchy may be subject to parameterization across languages (Scott 2002, p. 110).

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Cinque, in the ANA languages he considers (i.e., Romance), "the unmarked relative ordering of APs among each other is exactly the same as that of AN languages" (Cinque 1994, p. 29). Assuming that these languages start with an $A_1A_2A_3N$ structure (e.g., *pretty big red ball*), then by moving N above A_3 the preservation of the relative order among the adjectives follows: $A_1A_2NA_3$ (e.g., *pretty big ball red*).

In response to the observation that some postnominal adjectives in Romance in fact exhibit mirror-image ordering, Cinque takes such adjectives "not to be attributive, but predicative, as such outside the ordering restrictions holding of attributive APs" (Cinque 1994, p. 29). In other words, Cinque makes use of two strategies for introducing adnominal adjectives into the syntax: prenominally, via the specifier position of functional structure, and postnominally, via some method of predication. Only the former strategy attends to ordering considerations, and the structural details of the latter strategy are left unspecified (but see Cinque 2010, 2014). Laenzlinger (2005) rejects this analysis, although not the assumptions regarding a surfeit of functional structure. For Laenzlinger, all adjectives get introduced prenominally in the specifier position of the relevant functional projection, and the relative order of those projections is fixed cross-linguistically. However, different movement possibilities yield distinct surface orders. In prenominal languages like English, the absence of movement has the surface order directly reflect the hierarchy of functional projections (A_1A_2N). In postnominal languages displaying mirror-image preferences, a snowballing roll-up movement first has NP move to the specifier position of the closest functional projection, then that entire functional projection raises to the specifier of the next highest functional projection, and so on, resulting in a reverse of the word order (NA_2A_1). Another option has NP raise cyclically through the various functional projections, which preserves the ordering hierarchy postnominally (NA_1A_2). For Laenzlinger (2005), a Romance language like French makes use of either snowballing FP movement or cyclic NP raising, thereby delivering both NA_2A_1 and NA_1A_2 orderings.

2.4. Information-Theoretic Predictors

Recently, several authors have proposed that adjective ordering preferences fall out from more general efficiency pressures shaping constituent order across languages. Under these approaches, the considerations that determine the relative order of adjectives in multi-adjective strings arise whenever ordering decisions get made. The particular proposals find inspiration from information theory (Shannon 1948, Pierce 1980, Cover & Thomas 2006), which concerns the efficiency of digital communication codes. In order to use information theory to explain adjective ordering preferences, one must extend the relevant concepts from digital codes to natural languages (see Futrell 2017 for discussion), but, as we shall see, the intuitions at play may be as old as the study of adjective ordering itself.

The central notion of information theory is entropy, the inverse of predictability. Thinking in terms of probability distributions, a distribution with high peaks describes a more predictable situation (i.e., the outcomes with higher probability are more likely); a peaked distribution has low entropy. Conversely, a flat probability distribution represents maximal uncertainty (i.e., no outcome is more likely than any other); such a distribution has high entropy. Entropy relates to information content, as quantified by surprisal: A sample drawn from a distribution with low entropy is likely to be less surprising—and thereby less informative—than a sample drawn from a distribution with high entropy; entropy measures the expectation of surprise.

For Dyer (2017, 2018), entropy plays a central role in the cost of integrating dependent constituents with their heads, and pressures toward minimizing cost influence constituent ordering. Building on the idea from Gibson (1998) that integration cost comprises the distance between a

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head and its dependent (i.e., how long the dependency must be maintained in working memory) plus the complexity of the integration itself, Dyer argues that integration complexity is determined by the entropy of the probability distribution of a dependent's heads. Because integration complexity is additive across intervening elements of a dependency, Dyer proposes that adjectives with lower integration complexity should appear closer to the modified noun. The result has adjectives that compose with a more restricted set of heads closer to the noun—an idea familiar from Sweet's (1898) notion of meaning specificity or Ziff's (1960) privilege of occurrence.

For a different information-theoretic implementation of meaning specificity, Westbury (2021) builds on the psychological notion of need probability from memory research (Anderson & Milson 1989). Westbury proposes that the linear order of adjectives reflects their likely need, with adjective meanings that are a priori more likely to be needed occurring earlier. Words with a more general, multipurpose meaning will be used across a wider range of contexts, which means that, on average, the adjective we are most likely to need will be the one that occurs in the most contexts. Westbury implements this notion of likely need as a form of meaning distance: The more distant an adjective meaning is from the average adjective meaning, the more specific its meaning must be, and therefore its a priori likely need across contexts will be lower. Using tools from natural language processing to construct word meanings as vectors of numbers (or points in a high-dimensional space), Westbury defines average adjective meaning as the vector obtained by averaging the vectors for the 1,000 most common adjectives. Meaning distance is then implemented as an adjective vector's cosine distance from the average vector; adjectives with greater meaning distance (i.e., adjectives with higher meaning specificity) appear closer to the modified noun.

Futrell (2017, 2019) proposes a different information-theoretic quantity in his account of the pressures shaping word order: pointwise mutual information (PMI). PMI measures the extent to which two words predict each other, a way of quantifying the information they have in common. According to Futrell, words in a dependency that predict each other should occur nearer together. In a multi-adjective string, this notion of information locality has adjectives with higher PMI with the modified noun occur closer to the noun. By estimating PMI on the basis of co-occurrence statistics in a corpus, adjectives and nouns have higher PMI to the extent that they occur together more often than would be expected on the basis of their individual occurrence probabilities. In this way, PMI quantifies Sweet's (1898) intuition of collocation frequency: Adjectives that frequently modify nouns should appear closer to those nouns. PMI also offers an operationalization for meaning closeness.

Dyer et al. (2021) propose yet another information-theoretic predictor of ordering preferences: information gain, a measure of uncertainty reduction on the basis of the evidence, both positive and negative, provided by the occurrence of an adjective. For Dyer et al., speakers greedily maximize information gain, which means that adjectives that contribute greater information gain appear earlier than adjectives with less information gain. And because the information gained depends on the information already present, information gain as a predictor behaves the same regardless of whether adjectives occur prenominally (AAN), postnominally (NAA), or both (ANA, as in many Romance languages); in each case, the claim is that information gain should be maximized, such that adjectives with greater information gain occur linearly earlier.

2.5. Unmarked Versus Marked Contexts

So far, our discussion has focused on proposals meant to account for the default, unmarked ordering of modificational adjectives. However, the story of adjective ordering concerns more than just the default cases. Nearly all of the proposals surveyed above note that nonstandard, marked

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orderings arise frequently—and when they do, they communicate nonstandard, marked meanings. For example, already in 1945 (more accurately 1937, when the paper was written), Whorf (1945, p. 5) noticed that the default order (for him, determined by adjective inherentness) "may be reversed to make a balanced contrast, but only by changing the normal stress pattern." He gives the example of *French pretty girl*, which may be produced with stress on *French* "if in contrast with, e.g., *Spanish pretty girl*" (Whorf 1945, p. 5). Hill (1958, p. 175) notes an additional feature of the phonology of nonstandard orderings, namely the "juncture," or pause, that occurs between adjectives that have been reversed from their standard order.

Martin (1970) shows that the prevalence of junctures tracks the strength of the ordering preference violated in an order reversal; in other words, junctures become more likely as the nonstandard order becomes more marked. For Martin, the junctures that accompany preferred order reversals signal the atypical syntax that justifies the reversal. He argues that in standard orderings, both adjectives modify the noun independently, so that a *small brown box* is a box that is both small and brown. With juncture, *brown*, *small box* has *brown* modify *small box* as a unit. Martin & Ferb (1973) comment on the contextual support required for this modification structure, specifically the appropriate contrasts involved, so that *brown*, *small box* would be felicitous in a context where a speaker wants to signal the brown box among several small boxes.

Some authors have gone so far as to say that contextual pressures governing contrast rule the day in all cases of adjective ordering, and what we perceive as default orderings in fact arise from aggregating across typical contexts. Oller & Sales (1969) argue that more limiting adjectives—adjectives that establish a greater limit on the set of potential referents in a given context—appear earlier in English, farther from the modified noun; Danks & Glucksberg (1971) and Danks & Schwenk (1972) arrive at the same conclusion, using the label "discriminative" for adjectives that rule out more potential referents. In the *brown*, *small box* context, *brown* does more to rule out potential referents (i.e., the nonbrown boxes) than *small* does (all boxes are small in the context), and so *brown* appears farther from the noun. With this flexibility in mind, Ney (1983, p. 100) concludes that "almost any order of adjectives seems to be possible depending on the intended meaning of the speaker or the situation in which the speaker frames an utterance."

3. EVALUATING PREDICTORS

Having surveyed many of the predictors advanced in the literature, we turn now to empirical tests of the predictors. But first, a word of caution: While there exist many empirical investigations of adjective ordering preferences, most of them with the aim of evaluating the performance of predictors discussed in the previous section, few of these investigations pit predictors against one another, performing the sort of model comparison necessary to evaluate the relative performance of a predictor. Because of the diversity of data sets, dependent measures, and adjectives involved across these studies, it is often inappropriate to compare the performance of a predictor in one study with the performance of a predictor in another study. As a result, readers with an interest in which predictor outperforms the rest will no doubt be disappointed by the discussion that follows. Rather than crowning a victor, the goal is more modest: a survey of the empirical evidence in support of or against the predictors that have been advanced.

It should be clear that all investigations of adjective ordering are empirical in the sense that they engage with speaker intuitions or language data, sometimes from several languages. In an effort to streamline the discussion, here we focus on quantitative evaluations using behavioral and corpus data. Rather than structuring the discussion around individual studies, we consider each predictor in light of the evidence both for and against it. However, before turning to the specific predictors, two studies deserve special mention.

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Martin (1969a) performed some of the first behavioral studies on adjective ordering, and his work offers important methodological lessons for the studies that follow. In order to evaluate the predictions of his predictor, definiteness of denotation, Martin needs two ingredients: (*a*) a reliable estimate of the ordering preferences to be predicted, which he assesses quantitatively in a series of behavioral experiments eliciting preference judgments; and (*b*) a clear operationalization of his predictor so that it can be estimated from human judgments. Additionally, Martin (1969a) provides an alternative operationalization in terms of "absoluteness" (i.e., given a set of objects, how many would have to be compared before it would be possible to describe an object as ADJECTIVE?); he also compares his predictor against alternatives (i.e., imageability, inherentness, and frequency).

A number of behavioral studies followed Martin's in the 1970s; then there was a lull in quantitative investigations as class-based descriptions gained in popularity. Quantitative investigations resurfaced with the arrival of the new millennium, as Wulff (2003) pioneered large-scale corpus methodology applied to adjective ordering. Rather than explaining preferences obtained from controlled behavioral studies, Wulff set for herself a perhaps even more ambitious goal: explaining the language behavior speakers produce in the wild. Wulff (2003) also provides an early example of a multifactor approach, looking at the combined performance of several predictors.

3.1. Specific Predictors

Several studies have provided evidence in favor of closeness in meaning/the role of collocations using various operationalizations of the predictor (behavioral: Lockhart & Martin 1969, Martin 1969b, Byrne 1979, Westbury 2021; corpus: Wulff 2003, Hahn et al. 2018, Futrell 2019, Kotowski & Härtl 2019, Westbury 2021). The evidence against closeness in meaning as a predictor comes from behavioral studies that fail to find noun effects on ordering preferences (Martin & Ferb 1973, Scontras et al. 2017). All of these studies have investigated ordering preferences in English.

The picture for meaning specificity is more mixed. In her corpus analysis of English, Wulff (2003) fails to find support for meaning specificity, which she operationalizes as the number of different head nouns an adjective modifies. However, in his corpus analysis of at least 30 languages (including English), Dyer (2017) finds that meaning specificity, as operationalized by integration cost, successfully predicts a modest number of adjective orderings; in his words, "There seems to be an effect here, but it is not necessarily very strong" (Dyer 2017, p. 125; see Futrell et al. 2020 for a similar conclusion specific to English). Operationalizing meaning specificity as likely need, Westbury (2021) finds support from both corpus and behavioral evidence in English.

With inherentness, the evidence is less ambiguous—and also less encouraging. Using a behavioral operationalization, Martin (1969a) fails to find support for inherentness as a reliable predictor in English; Scontras et al. (2017) find the same, using Martin's operationalization. Kotowski & Härtl (2019) focus on the temporariness dimension of inherentness and fail to find an effect in their corpus analysis of German; Dyer (2017, p. 11) arrives at the same conclusion on the basis of a corpus investigation of English.

Martin (1969a) finds clear behavioral support for definiteness of denotation and its operationalization via absoluteness. Focusing just on absoluteness, Wulff (2003) reaches the same conclusion using corpus data. Both studies limit their focus to English.

A number of studies have found support for subjectivity as a predictor of English ordering preferences (behavioral: Scontras et al. 2017; corpus: Wulff 2003, Hill 2012, Scontras et al. 2017, Hahn et al. 2018, Futrell et al. 2020). The behavioral methodology from Scontras et al. (2017) has been extended to document subjectivity-based ordering preferences in Mandarin (Shi & Scontras 2020), Arabic and Heritage Arabic (Kachakeche & Scontras 2020), Spanish (Scontras et al. 2020), and Greek, Hebrew, and Vietnamese (Scontras et al. 2021). Relying on English estimates of

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subjectivity, Trainin & Shetreet (2021) find general support for subjectivity as a predictor of ordering in Hebrew (see also Trainin & Shetreet 2022), although two of the adjective classes they investigated in their behavioral experiments (pattern and texture adjectives) behaved unexpectedly from the perspective of subjectivity alone. We return to this finding in our discussion of preference strength and dependent measures in Section 3.2.

I am unaware of any studies testing derivational syntactic proximity as a predictor, although work investigating the brain responses to nonstandard orderings finds that order violations elicit ERP components typical of unexpected semantic content (i.e., larger N400s) rather than of unexpected structure (i.e., P600s) (Huang & Federmeier 2012). In other words, order violations implicate heightened processing of adjective meaning rather than structure.

Studies that have investigated nounlikeness report conflicting results. In her analysis of the spoken portion of the British National Corpus, Wulff (2003) finds a small effect of nounlikeness, but in the opposite direction of what was predicted by Bever (1970): More nounlike adjectives were slightly more likely to occur farther from the noun. In a separate corpus analysis of English using the much larger Google Ngram corpus, Hill (2012) finds that more nounlike adjectives occur closer to the noun, although again the effect is rather small. Both authors operationalize nounlikeness in the same way, so the source of the difference in the findings remains a mystery.

Investigations of adjective frequency have consistently found that more frequent adjectives tend to occur before less frequent adjectives, although the studies have limited their focus to English (behavioral: Martin 1969a, Scontras et al. 2017; corpus: Wulff 2003, Scontras et al. 2017, Trotzke & Wittenberg 2019, Westbury 2021). With adjective length, Wulff (2003) and Scontras et al. (2017) show it to be a reliable predictor of English ordering, although Kotowski & Härtl (2019) arrive at the opposite conclusion in their corpus analysis of German.

Wulff (2003) finds evidence in support of Dixon's (1977) lexical class hierarchy in English, although she ultimately concludes that her operationalization of subjectivity subsumes Dixon's hierarchy. Bar-Sever et al. (2018) use English corpus evidence to document ordering preferences consistent with a class hierarchy in children 4 years of age (but not younger). Evidence against the strong version of the lexical class hypothesis comes from the repeated general observation that ordering preferences are gradient, much weaker than a full-fledged prohibition on ungrammatical orderings would predict (e.g., Danks & Glucksberg 1971, Scontras et al. 2017, Leivada & Westergaard 2019); these gradient preferences result in productions far more variable than would be allowed by a strict hierarchy (e.g., Truswell 2009, Kotowski & Härtl 2019). The brain data from Huang & Federmeier (2012) mentioned above in the context of derivational syntactic complexity also appear to be inconsistent with a purely syntactic account based on functional projections.

Leung et al. (2020) use a different approach to assessing class-based hierarchies. Rather than taking a specific hierarchy off the shelf and evaluating its performance, the authors use large-scale cross-linguistic corpus evidence to both infer the relevant classes (they have their models infer 15 totally ordered lexical classes, more than double many of the existing hierarchies) and predict attested orderings using those classes.[8] The model succeeds on English. More importantly, the model can transfer its knowledge cross-linguistically, using a class hierarchy learned from a subset of languages to predict ordering in 24 unseen languages. With the possibility of class-based cross-linguistic transfer, Leung et al. provide perhaps the most comprehensive evidence in favor of mirror-image preferences pre- versus postnominally. But before concluding that Leung

[8]The authors provide anecdotal evidence suggesting that their inferred classes are compatible with those of Cinque (2010), but Cinque only specifies the ordering of five classes (value > size > shape > color > provenance).

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et al. provide strong evidence in favor of lexical classes, we must recognize that the inferred 15-class hierarchy may simply approximate one of the other predictors [as Wulff (2003) concluded in her study].

Only one study has looked at information gain, but its scope dwarfs many of the previous investigations. Dyer et al. (2021) investigate information gain using corpora of 32 languages; of these, information gain successfully predicts attested orderings in 29 languages. Notably, Dyer et al. test the predictions of information gain in three different ordering templates: AAN, NAA, and ANA—the only study mentioned in this section to look at mixed ANA orderings.

Danks & Schwenk (1972) manipulate the communication context in a production study of English, finding that more referentially limiting or discriminating adjectives tend to occur earlier than less discriminating adjectives (see also Fukumura 2018). Fukumura & Santesteban (2017) report results from Basque, a language with postnominal adjectives, showing that speakers also place more discriminating adjectives earlier.

3.2. Dependent Measures and Preference Strength

The discussion above has largely ignored the methodological details of the studies meant to argue for or against various predictors: the specific adjectives tested, the dependent measure, and so on. However, methodology matters for the results that are attained. Take the case of Greek. Leivada & Westergaard (2019) report the results of a three-point Likert-scale acceptability-rating study, looking at order permutations in three pairings of adjective classes (size–nationality, color–shape, and subjective comment–material). Regardless of the order of adjectives, by far the most common response is that the orderings are "correct" (as opposed to "neither correct nor wrong" or "wrong"). The authors interpret this finding as evidence against hard-coded grammatical restrictions on adjective ordering in Greek, arguing against a class-based hierarchy. Without a significant difference in the acceptability of ostensible ordering violations across pairs of adjective classes, one may be tempted to interpret these results also as evidence against ordering preferences in Greek—or at least weaker preferences in Greek than in often-studied English.

Here is where the different methods become relevant. In his corpus study, Dyer (2017, p. 124) shows integration cost to be a modest predictor of ordering preferences in Greek, thereby implying that there are preferences to predict in the language. Moreover, integration cost performs better in Greek than it does in English. We find a similar picture in Dyer et al.'s (2021) corpus study, where information gain is shown to predict adjective ordering in both Greek and English—and to a similar extent in the two languages. Behaviorally, Scontras et al. (2021) use the preference-inference methodology from Scontras et al. (2017) to document clear ordering preferences in Greek (that happen to be predicted by subjectivity); but although the authors do not compare English and Greek directly, it appears to be the case that the Greek preferences are slightly weaker than those measured for English using the same methodology.

It would seem that rating the acceptability of a multi-adjective nominal may be a less sensitive measure for picking up on preferences than using large-scale corpus counts or rating the relative naturalness of two competing orderings; but there still remains the possibility that different languages have preferences of different strengths. Unfortunately, few studies have been designed with the goal of comparing preference strength across languages. One exception is provided by Trainin & Shetreet (2021), who compare Hebrew and English. The authors use three different methodologies: free production, naturalness ratings of isolated orderings, and forced choice between two competing orderings. Across all three paradigms, the authors find that preferences in Hebrew are weaker or less categorical than those in English. Moreover, both Dyer (2017) and Dyer et al. (2021) find that their predictors—integration cost and information gain, respectively—at least

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trend such that they predict less well in Hebrew than in English. Given that Hebrew has postnominal adjectives while English adjectives are prenominal, we may wonder whether the different communicative status of pre- versus postnominal adjectives may affect the strength of ordering preferences.[9] However, Dyer et al. (2021) assess the relative flexibility of adjective ordering in 18 prenominal versus 13 postnominal languages and find that the rate of ordering flexibility (calculated as the number of adjective pairs attested in both possible orders) is comparable across templates. But, as the discussion above highlights, different measures may be more or less sensitive as assessments of a given research question regarding adjective ordering. For assessing relative preference strength, additional cross-linguistic behavioral work may be necessary.

3.3. Summary

Taking stock, the empirical picture of adjective ordering across languages aligns with the predictions of many of the proposed accounts. Only inherentness lacks independent evidence. Still, many avenues remain unexplored. Unsurprisingly, most investigations have focused on English, which means many predictors have yet to be vetted cross-linguistically.[10] Corpus work supporting stable preferences cross-linguistically suggests that predictors that are successful in English may find similar success in other languages, but we should temper our confidence with the reality that language data rarely fail to surprise. Perhaps the most glaring hole in the empirical landscape is cross-linguistic behavioral work, which is lacking for all of the predictors except for subjectivity and lexical class hierarchies. Also lacking are studies explicitly comparing ordering preferences—and, crucially, the strength of such preferences—across languages. Finally, despite a few notable exceptions (e.g., Martin 1970, Danks & Glucksberg 1971, Trainin & Shetreet 2021), nearly all of the behavioral work has focused on preferences for multi-adjective strings with only two adjectives.

4. EXPLAINING PREDICTORS

With a set of candidate predictors and some sense of the evidence both for and against them, the last order of business concerns explanations for why a predictor should play the role it does in adjective ordering. We begin with a note on grammatically encoded class-based hierarchies as explanations of adjective ordering, then we turn to the pressures—both psychological and communicative—that may be at play in explaining why certain predictors succeed.

4.1. Class-Based Descriptions

I have taken care to describe proposals appealing to lexical classes as "descriptions" because their focus lies squarely on mapping out the ordering regularities and providing an architecture within which to formalize that mapping—whence the self-ascribed name for the enterprise, "cartography" (cf. Cinque & Rizzi 2008). However, these proposals are not devoid of explanatory power. By attributing ordering regularities to Universal Grammar, the proposals offer a ready answer to the question of why we should find similar preferences cross-linguistically: The functional hierarchy that determines adjective order comes prespecified as part of being human. Still, these proposals have very little to say about why the specific hierarchy we find is the one Universal Grammar

[9]Prenominal adjectives may be more useful for the purpose of incremental reference resolution, which results in different usage rates (Rubio-Fernández 2016, Kachakeche et al. 2021).

[10]I have also focused the discussion on investigations of adult preferences. For more on the development of ordering preferences in children, readers are referred to Bever (1970), Scheffelin (1971), Martin & Molfese (1971, 1972), Ford & Olson (1975), Hare & Otto (1978), and Bar-Sever et al. (2018).

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has settled on and not some other hierarchy—is it merely an accident of being human that size adjectives appear before color adjectives prenominally?

Recently, Larson (2021) revisited the cartographic enterprise with the aim of increasing its explanatory power while sidestepping some of its shortcomings. Finding inspiration from mathematics and the linear ordering of numbers, Larson reimagines the architecture so that a single relation orders the (features corresponding to) different lexical classes. Inspired by the experimental results of Scontras et al. (2017), he proposes that subjectivity determines this relation between classes. Thus, instead of stipulating that, say, the functional head handling size adjectives functionally selects the functional projection handling length, Larson orders the feature associated with size adjectives above all other features associated with lexical classes with lower subjectivity (including, according to Larson, length adjectives).

It would seem, then, that even if speakers represent ordering preferences via articulated functional hierarchies, some other factor(s) must be at play in delivering the specific orderings observed. A full explanation of adjective ordering preferences will need to address those factors.

4.2. Psychological and Communicative Pressures

Sticking with the map-making analogy, a cartographic description calls for a geological explanation of the pressures giving rise to a structure. Here we consider five such pressures.

When dealing with questions of order, considerations of accessibility frequently arise: The first element may simply be the easiest one to produce in the moment (e.g., Bock 1982, 1987; Ferreira & Dell 2000). Indeed, Martin (1969a) raises accessibility as the pressure shaping adjective order. Perhaps differential meaning closeness—or specificity or subjectivity or discriminability—yields differential accessibility. From the perspective of linear order, it makes sense that more accessible things would get accessed, and therefore appear, earlier. The evidence from frequency supports this idea: More frequent adjectives occur earlier, at least according to the evidence from English. Length may behave similarly, such that shorter, lighter-weight adjectives are more easily accessed and therefore occur earlier. For Martin, accessibility also explains the role of absoluteness and definiteness of denotation in ordering preferences: It should take different amounts of time for a speaker to evaluate the application of less versus more context-sensitive properties requiring fewer versus more comparisons in their evaluation. One may attempt to tell similar stories for other predictors.

The trouble with accessibility as the primary factor in ordering preferences concerns mirror-image preferences cross-linguistically: We would need accessibility to operate in the opposite direction postnominally such that more accessible adjectives occur later. Martin (1969b) attempts to skirt this problem by stipulating that prenominal adjectives derive from postnominal base positions, such that *small brown box* begins as *box brown small* before transformations apply. But if accessibility drives ordering preferences, and postnominal base positions prefigure prenominal multi-adjective strings, then more frequent adjectives should appear closer to the noun regardless of whether the adjectives are pre- or postnominal. In other words, in English, more frequent adjectives should appear later, contrary to what has been observed in several studies, including Martin's (1969a) own. It therefore seems unlikely that accessibility serves as a general-purpose pressure shaping prenominal orders and their mirror image in postnominal languages, although accessibility no doubt plays a role in availability-based production.

Rather than attributing ordering preferences to the online pressures a speaker faces when accessing words, Bever (1970) invokes pressures introduced by the human perceptual system. In his discussion of the segmentation of noun phrases with prenominal modifiers, Bever proposes that the parser attends to changes in nounlikeness in an effort to identify constituent boundaries.

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According to Bever, the presence of a determiner identifies the beginning of a noun phrase, while a drop in nounlikeness signals its end. In a phrase like *the brown big box*, a linear parser might mistakenly identify the drop in nounlikeness from *brown* to *big* with a phrase boundary; with the preferred ordering, nounlikeness drops after the noun (*the big brown box weighs a ton*). While the parsing story for nounlikeness is clear under this view, it may also be possible to tell similar stories for some of the other predictors: Perhaps more definite, absolute, objective properties serve more often as nouns. Ignoring the difficulty of extending this analysis to languages with postnominal or mixed (i.e., ANA) adjectives, the trouble with this parsing story as a general account of ordering preferences lies in the conflicting empirical picture considered in the previous section: To the extent that nounlikeness behaves as predicted by Bever (1970), the effect is modest.

While ordering preferences may not arise solely in an effort to relieve the burden of the parser, the preferences may still represent the output of an optimization procedure. For Dyer (2017, 2018) and Futrell (2019), ordering preferences yield structures that minimize the information-processing costs of comprehenders interpreting those structures.[11] Dyer quantifies processing cost using integration complexity, which implements Sweet's (1898) idea of meaning specificity; by placing adjectives with lower integration cost nearer to the noun, comprehenders have an easier time integrating dependent constituents with their heads. For Futrell (2019), processing cost decreases with increasing prediction accuracy for the unfolding of words in linear time; assuming that comprehenders have finite memory resources, placing two elements that predict each other (i.e., with high PMI or collocation frequency) nearer to one another maximizes prediction accuracy and therefore minimizes processing cost.

Beyond efficiency in language processing, some authors have considered communicative success as a driving factor in ordering preferences: Perhaps certain orderings are more useful at getting across the intended meaning. Recently, this pressure has been explored in explanations of subjectivity-based ordering preferences. However, when it comes to fleshing out the story, there are several choice points. Existing accounts agree on the idea that less subjective content is more useful at successfully communicating a message, given that listeners will be more likely to arrive at an interpretation that aligns with that of the speaker. For example, a listener and speaker are more likely to agree on the set of *brown* things than they are to agree on the set of *small* ones. Hahn et al. (2018) identify the fallibility of memory representations as the primary mechanism incentivizing the placement of less subjective, more useful content nearer to the noun: Better to have that more useful content close at hand (i.e., less susceptible to corruption by noisy memory) when arriving at the interpretation of a nominal phrase.

An alternative view links subjectivity to the hierarchical structure of restrictive modification, with adjectives linearly closer to the noun making their semantic contribution—restricting the nominal denotation—earlier than adjectives farther away (Simonič 2018, Franke et al. 2019, Scontras et al. 2019). A noun like *box* will identify a set of potential referents; if the goal is to reference a single box among this set—say the small brown one—a listener requires additional information about precisely which box is under consideration. By adding an adjective, we restrict the set of potential referents: *Brown* limits the set to just those boxes that are brown, and *small* limits that set to just the brown boxes that are small. When several adjectives are necessary, the less subjective adjectives should be used to restrict a larger set (i.e., less subjective adjectives should make their semantic contribution earlier), since with larger sets of potential referents there will be more opportunities for miscommunication: For each potential referent, a listener must decide

[11]While these accounts focus on the processing burden of comprehenders, it seems plausible that speakers benefit from similar pressures operating during production.

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whether the relevant property holds, and with subjective properties there is potential for faultless disagreement between a speaker and a listener.

Scontras et al. (2020) provide an empirical argument in favor of the incremental restriction account over a memory-based one, using data from conjunction. As reviewed above, memory pressures favor placing less subjective adjectives linearly closer to the noun, regardless of structure, in an effort to avoid corruption of the memory representation of more useful, less subjective content. With incremental restriction, the pressure arises when certain adjectives make their semantic contribution earlier than others; in such a case, communication will be more successful when less subjective adjectives make their contribution earlier. But when adjectives restrict the set of potential referents together, at the same time, the pressure for subjectivity-based ordering no longer arises according to incremental restriction. For Scontras et al. (2020), conjunction presents such a situation: In *small and brown box*, the adjectives first form a constituent to the exclusion of the noun, then that constituent restricts the nominal denotation. Crucially, the presence of conjunction also weakens adjective ordering preferences (Ford & Olson 1975, Byrne 1979, Rosales & Scontras 2019, Scontras et al. 2020; see also Hetzron 1978, p. 174).

The pressures considered so far have assumed genuine optionality in adjective ordering: Given an intended meaning, which ordering yields the most efficient delivery device? However, different orderings may have different meanings (see the discussion in Section 2.5 above), and the choice of ordering may come down to a choice of the meaning one wants to express. In other words, what we have considered as ordering preferences may in fact represent speaker decisions about intended meanings—a ready explanation for discriminability as a predictor, but also perhaps for meaning closeness, meaning specialization, or nounlikeness. The trick lies in characterizing ordering regularities across contexts so that a series of individual meaning-based ordering decisions result in general tendencies. For Danks & Glucksberg (1971, p. 67), what speakers perceive as general preferences simply represent how adjectives get used in "their most usual or frequent context." However, while it may be the case that meaning nuances influence individual orders, it seems unlikely that they will explain the full range of observed preferences; as Ney (1983, p. 101) observes, "semantic criteria do not dominate in the selection of the order of adjectives."

5. LOOKING FORWARD

After over a century of work on adjective ordering, many questions—both empirical and theoretical—remain unanswered. However, when it comes to the source of ordering preferences, the picture appears to be relatively clear: "There is no single and all-encompassing answer to the question of how adjectives are ordered linearly in all real speech situations or written texts" (Hetzron 1978, p. 178). In other words, there is no single source of ordering preferences. Rather, the regularities that we witness represent the result of several pressures, some of them conflicting. Considerations of accessibility, processing cost, communicative success, and choice of meaning are all likely to influence what we perceive as ordering preferences—and no doubt there are other considerations yet to be examined.

As research on adjective ordering continues through its second century, progress will be made as researchers embrace the complexity of interacting factors at play in the determination of adjective order. Indeed, some researchers have already arrived at a multifactor approach—at least in English. Wulff (2003) uses several predictors to capture noisy naturalistic productions. Futrell et al. (2020) find that a two-factor account incorporating information about adjective meaning (operationalized by subjectivity or information gain) and collocation frequency (operationalized by PMI) outperforms any single predictor. The work now lies in extending the empirical reach of such approaches cross-linguistically, in an attempt to better understand both the regularities and differences across languages.

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With a clearer picture of the many pressures at play, we can make progress on a perhaps more elusive question: How much of what we observe as ordering preferences derives from online considerations as speakers communicate with each other rather than from a replication of regularities already present in the input? In their accounts of subjectivity-based preferences, Franke et al. (2019) and Scontras et al. (2019) raise the possibility that the preferences gradually emerge as speakers slowly attend to the communicative successes and failures that result from their utterances; those orderings that yield success are more fit evolutionarily, and so they are more likely to stick around. It seems plausible that a similar story could be told for the other efficiency-based predictors: Less costly, easier-to-produce orderings are more likely to proliferate. Viewed on an evolutionary timescale, ordering decisions may require very little attention to online considerations; the slightest benefit for one ordering over another can calcify over time into a robust preference. Importantly, by gaining clarity on the nature of ordering preferences in real-time language use, we arrive at a better understanding of the knowledge that constitutes them.

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